

Strengthening the accumbal indirect pathway promotes resilience to compulsive cocaine use

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A hallmark of addiction is the loss of control over drug intake, which is seen in only a fraction of those exposed to stimulant drugs such as cocaine. The cellular mechanisms underlying vulnerability or resistance to compulsive drug use remain unknown. We found that individual variability in the development of highly motivated and perseverative behavior toward cocaine is associated with synaptic plasticity in medium spiny neurons expressing dopamine D2 receptors (D2-MSNs) in the nucleus accumbens (NAc) of mice. Potentiation of glutamatergic inputs onto indirect pathway D2-MSNs was associated with resilience toward compulsive cocaine seeking. Inhibition of D2-MSNs using a chemogenetic approach enhanced the motivation to obtain cocaine, whereas optogenetic activation of D2-MSNs suppressed cocaine self-administration. These results indicate that recruitment of D2-MSNs in NAc functions to restrain cocaine self-administration and serves as a natural protective mechanism in drug-exposed individuals.

Individuals suffering from addiction endure large personal and financial losses to maintain drug use. Among other addictive behaviors, they show a strong perseverance and an extraordinary motivation to obtain the drug. These behaviors are expressed by only a fraction of those exposed to the drug, revealing a substantial degree of individual variability and the existence of predisposing traits and conditions that may serve as risk or protective factors in the development of addiction. In humans, the vulnerability to develop compulsive behaviors toward stimulant drugs has been linked to deficits in cortico-striatal processing and low levels of dopamine D2 receptors in the striatum^{1–3}. Moreover, impulsivity traits and low levels of dopamine D2 receptors have been associated with compulsive cocaine use in both rodents and nonhuman primates^{4,5}. Furthermore, rodents also show natural individual variability in the motivational properties of cocaine and the development of compulsive behaviors^{6–8}.

Dopamine D2 receptors are expressed in the subpopulation of medium spiny neurons (D2-MSNs) in the striatum that form indirect projections to midbrain regions via pallidum and subthalamic nuclei (indirect pathway). The other subpopulation of MSNs expresses dopamine D1 receptors (D1-MSNs) and forms direct projections to midbrain neurons (direct pathway). Activation of dopamine receptors on each subpopulation of MSNs triggers different intracellular signaling cascades. Activation of D2 receptors inhibits PKA activity via G_i signaling in D2-MSNs, whereas activation of D1 receptors stimulates PKA activity via G_{s/olf} signaling in D1-MSNs⁹. It is thought that these two MSN subtypes and their parallel pathways exert complementary, and sometimes opposing, actions on behaviors that are controlled by the cortico-striatal system¹⁰. The use of pharmacological tools that target D1 and D2 receptors has helped to determine the relative

contribution of the direct and indirect pathways in behavior. However, the complex expression pattern of dopamine D2 receptors present in both pre- and postsynaptic compartments in different neuronal types in the mesolimbic circuit has complicated the interpretation of these experimental results.

Cell type-specific approaches have recently been used to aid in this quest. In the dorsal striatum, optogenetic activation of direct pathway D1-MSNs increases locomotion and serves as a reinforcer, whereas activation of indirect pathway D2-MSN increases freezing behaviors and is not a reinforcer^{11,12}. In the NAc, a region involved in cue-induced reward learning, D1-MSNs and D2-MSNs have opposite effects on cocaine-related behaviors⁹. Activation of D2-MSNs reduces conditioned place preference for cocaine, whereas activation of D1-MSN increases it¹³. In addition, the ablation or inhibition of D2-MSNs in the NAc induces an increase in amphetamine conditioned place preference and facilitates locomotor sensitization to cocaine, suggesting a tonic role of D2-MSNs on limiting the actions of stimulant drugs^{14,15}.

However, despite these findings, the role of the indirect pathway and D2-MSNs in voluntary cocaine self-administration and compulsive drug seeking remains unclear. We predicted that indirect-pathway D2-MSNs would exert an inhibitory influence on behavioral output of this circuitry and limit drug seeking, and that weakening this pathway would remove the inhibitory control and render individuals more susceptible to develop compulsive drug seeking. We found individual variability in vulnerability to compulsive cocaine use that was correlated to the synaptic strength of inputs to D2-MSNs. Moreover, inhibition or activation of the accumbal indirect pathway enhanced or suppressed cocaine self-administration behavior, respectively.

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RESULTS

Individual variability in behaviors toward cocaine

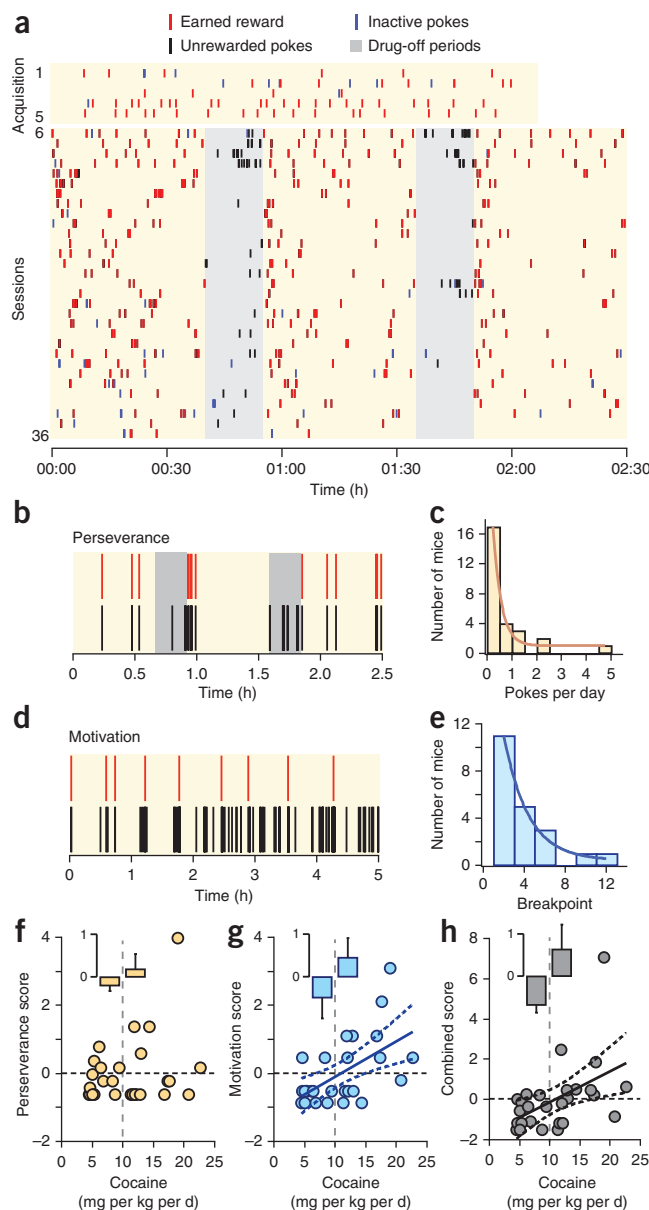
Intravenous cocaine self-administration was established in an outbred strain of mice using a cued-operant task that required naive mice to nose poke in an active hole to earn an intravenous infusion of cocaine (Supplementary Fig. 1). Approximately 55% of mice acquired the behavior within 5–10 d and the rest were removed from the study. Mice were then given access to cocaine during daily 2-h sessions for 6–7 weeks. Two behaviors were measured to determine the degree of compulsive drug use: the difficulty of stopping or limiting drug use, measured as perseverative drug seeking, and high motivation to obtain the drug, measured as the effort exerted to obtain it. These behaviors were adopted from the diagnostic criteria for drug dependence in humans described in the DSM-IV and have been successfully applied in a model developed in rats^{4,7,16}.

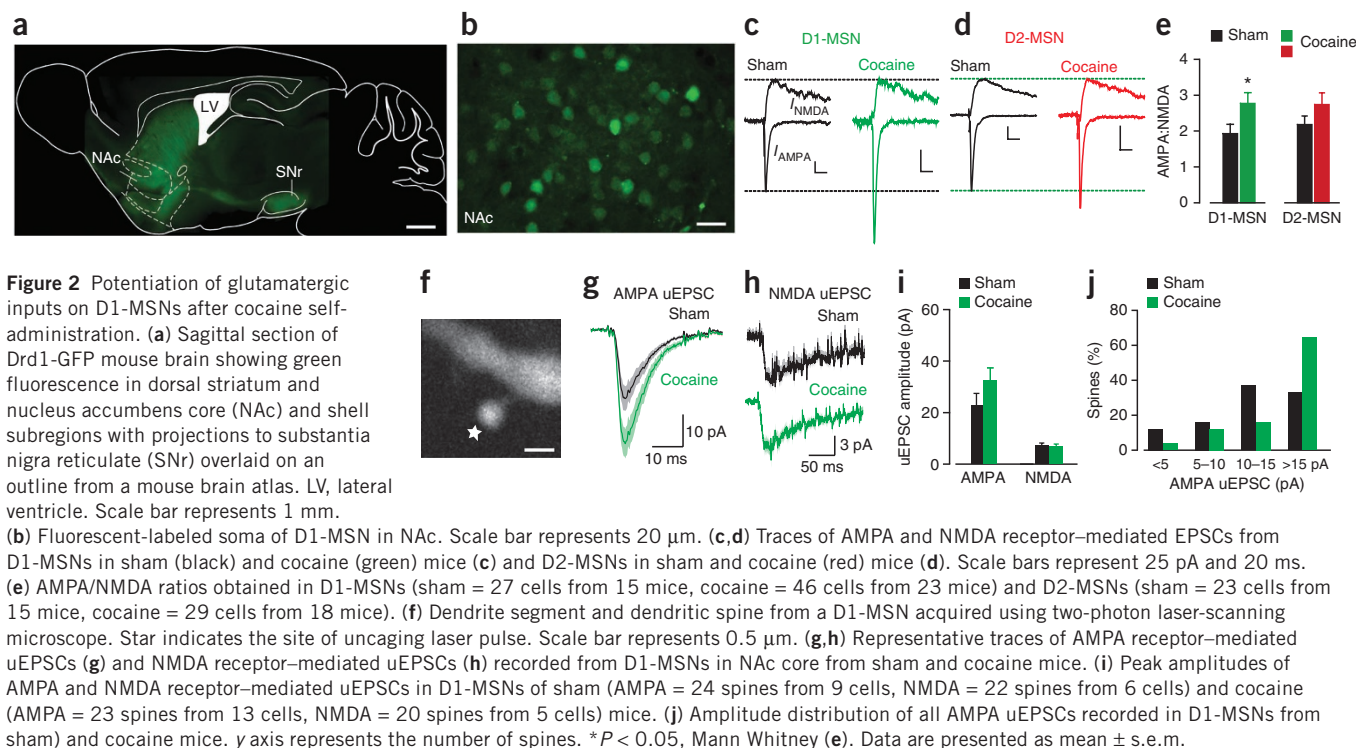
To measure perseverative responding, we interrupted sessions with two drug-off periods (15 min long), during which cocaine was not available (Fig. 1a–c). Initially, mice showed a typical extinction response during the drug-off periods that was characterized by a spike in responding on days 1 and 2, followed by a decrease in responding selectively in the drug-off periods during subsequent days. Although most mice maintained minimal responding for the rest of the sessions, a few developed perseverative responding after weeks of self-administration (Supplementary Fig. 2). The perseverance value was calculated as mean active pokes during the drug-off periods over the last ten sessions (Fig. 1c). During this time, perseverative responding during the drug-off period was selective to the active nose pokes and the number of inactive pokes was similar between mice with high and low perseverance values (0.13 ± 0.07 inactive pokes per day for mice with perseverance value > 2 , $n = 3$ mice; 0.27 ± 0.08 inactive per day for mice with values < 2 , $n = 24$ mice; unpaired t test, $t_{24} = 0.5$, $P = 0.62$).

Motivation to obtain cocaine was assessed in two progressive ratio sessions in which the number of pokes required to earn each consecutive cocaine reward was increased exponentially (Online Methods and Fig. 1d,e). The breakpoint, the number of pokes performed to earn the last reward, was defined as the motivation value. Similar to the measure of perseverative behavior, most mice showed low motivation values, but a few mice had high values (Supplementary Fig. 2). Notably, the ratio of inactive to active pokes did not change during these progressive ratio sessions (session, 0.27 ± 0.04 ; progressive ratio 0.48 ± 0.17 ; paired t test, $t_{26} = 1.2$, $P = 0.23$, $n = 27$ mice).

Figure 1 Individual variability in the behaviors toward cocaine after extended self-administration. (a) Raster plot showing the daily pattern of cocaine self-administration of a mouse during acquisition (day 1–5) and sessions (day 6–36). (b) Example of perseverative responding during the drug-off periods in session 35; red ticks indicate cocaine infusion, black ticks indicate active poke. (c) Distribution of the perseverance value (mean active pokes per d during drug-off) in the population. Line is log normal fit to the data (with $y_0 = 16.5$, $A = -14$, $x_0 = 1.5$, width = 4, $\chi^2 = 22.9$, $n = 10$ bins). (d) Example of responding during progressive ratio session used to measure motivation. The number of active pokes (black ticks) required to earn a cocaine infusion (red ticks) increased with each subsequent infusion. (e) Distribution of the breakpoint values in the population. Line is log normal fit to the data (with $y_0 = 15.5$, $A = -9$, $x_0 = 10.8$, width = 2.2, $\chi^2 = 1.7$, $n = 6$ bins). (f–h) The scores of individual mice for perseverance (f), motivation (g) and combined behavioral scores (h) versus cocaine intake. The vertical dashed lines mark the 10 mg per kg per d cocaine dose and the horizontal dashed lines mark the mean of score (zero). Insets, mean $\pm$ s.e.m. perseverance score for mice with cocaine intake (left) lower and (right) higher than 10 mg per kg per d. Solid and dashed lines represent linear regression fit and 95% confidence interval.

A combined behavior score was calculated for each mouse by adding the z score of perseverance and motivation values. By definition, z scores have a mean of zero and a s.d. of 1, which ensures equal contribution of each behavior and eliminates inherent differences in the variance of each measurement. Mice varied on the degree of perseverance, motivation and combined behavior score, as well as on their average cocaine intake through the study (5–24 mg per kg of body weight per d). However, there was no correlation between the perseverance scores and cocaine intake ($r_p = 0.27$, $r^2 = 0.08$, $P = 0.16$, $n = 27$ mice; Fig. 1f). Although the highest perseverance score was observed in a mouse with a daily intake of 20 mg per kg of cocaine, high intake was not sufficient to predict positive perseverance scores. As expected, there was a positive correlation between the motivation and the combined behavior scores and daily cocaine intake (motivation score: $r_p = 0.57$, $r^2 = 0.31$, $P < 0.01$, $n = 27$ mice; combined score: $r_p = 0.49$, $r^2 = 0.24$, $P < 0.01$, $n = 27$ mice; Fig. 1g,h). A mean daily intake of 10 mg per kg per d, a dose that increases locomotion in this mouse strain¹⁷, was used as the threshold to identify mice with high and low cocaine intake. Mice with low daily cocaine intake displayed





mean negative motivation scores (-0.5 ± 0.1) and behavior scores (-0.7 ± 0.2 , $n = 13$ mice), whereas mice with high cocaine intake showed mean positive motivation (0.5 ± 0.3) and behavior scores (0.6 ± 0.6 , $n = 14$ mice) (Fig. 1g,h).

Despite the correlation between intake and the behavior scores, there was a considerable degree of variability in the scores among mice with high intake. High intake did not always result in the development of compulsive cocaine use, and some mice maintained negative behavior scores. We asked whether specific synaptic mechanisms are associated with and responsible for the vulnerability or resistance to develop compulsive behaviors among cocaine users.

Potentiation of excitatory synapses onto D1-MSNs

We examined glutamatergic excitatory transmission in direct-pathway D1-MSNs and indirect-pathway D2-MSNs in the NAc core region of *ex vivo* brain slices. These neurons received dense innervation from cortical regions that were activated by electrical stimulation. The expression of GFP was used to discriminate between fluorescently labeled D1-MSNs and unlabeled putative D2-MSNs (Fig. 2a,b). Recordings were made from 30 mice that underwent at least 17 sessions of cocaine self-administration (22 of these with behavioral scores) and 20 sham surgery mice. In general, we recorded from D1-MSNs in some mice and D2-MSNs in others, although recordings were obtained from both D1-MSNs and D2-MSNs in a small subset of cocaine-exposed mice (Supplementary Table 1). In cocaine-exposed mice, the ratio of AMPA/NMDA excitatory postsynaptic currents (EPSCs) was increased selectively in D1-MSNs, but not in D2-MSNs, compared with sham controls (D1-MSNs: cocaine, 2.77 ± 0.28 , $n = 46$ cells from 23 mice; sham, 1.93 ± 0.26 , $n = 27$ cells from 15 mice; Mann-Whitney test $U = 415$, $P = 0.018$; D2-MSNs: cocaine, 2.74 ± 0.31 , $n = 29$ cells from 18 mice; sham, 2.25 ± 0.27 , $n = 23$ cells from 15 mice, Mann-Whitney test $U = 312$, $P = 0.20$; Fig. 2c–e).

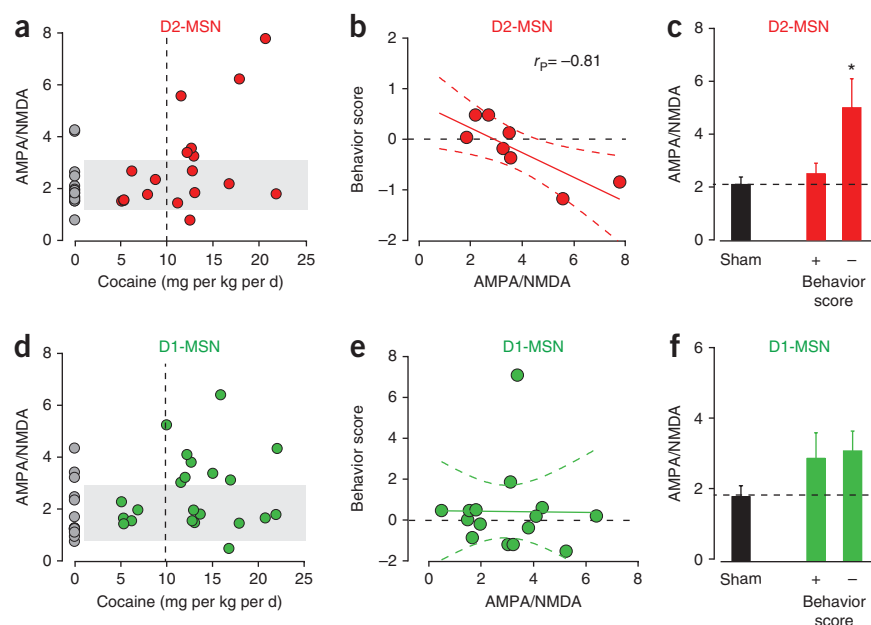
We further investigated this increase in the ratio of D1-MSNs using two-photon uncaging of glutamate at single spines to measure the

AMPA and NMDA receptor-mediated currents at individual spines (Fig. 2f–h). The amplitudes of AMPA receptor uncaging-evoked EPSCs (uEPSCs) in cocaine-exposed mice had a tendency to be larger than those in sham mice (sham, 23.3 ± 4.3 pA, $n = 24$ spines from 9 cells; cocaine, 32.9 ± 4.6 pA, $n = 23$ spines from 13 cells; Mann-Whitney test $U = 195$, $P = 0.09$ for two-tail and $P = 0.04$ for one-tail; Fig. 2i), whereas the amplitudes of NMDA receptor uEPSCs were similar in sham and cocaine-exposed mice (sham, 6.4 ± 0.7 pA, $n = 22$ spines from 6 cells; cocaine, 5.9 ± 0.8 pA, $n = 20$ spines from 5 cells; unpaired t test, $t_{40} = 0.26$, $P = 0.79$; Fig. 2i). A rightward shift in the amplitude distribution of AMPA receptor uEPSCs was seen in cocaine-exposed mice, indicating that there was a smaller percentage of spines with currents that were undistinguishable from noise (<5 pA uEPSC; cocaine, 4%; sham, 12.5%; Fig. 2j), and a greater percentage of spines with large AMPA receptor uEPSCs (>15 pA uEPSC; cocaine, 65%; sham, 33%; Supplementary Fig. 3c). Furthermore, we observed an increase in spine head width, but not length, in D1-MSNs after cocaine exposure (thin spines: sham, 0.81 ± 0.03 μ m, $n = 23$ spines; cocaine, 0.89 ± 0.03 μ m, $n = 22$ spines; wide spines: sham, 1.17 ± 0.04 μ m, $n = 23$ spines; cocaine, 1.32 ± 0.06 μ m, 22 spines; two-way ANOVA, no interaction, $F_{1,68} = 0.6$, $P = 0.4$; effect of width, $F_{1,68} = 99.6$, $P < 0.01$; effect of treatment, $F_{1,68} = 8.4$, $P < 0.05$; Bonferroni *post hoc* wide spines sham versus cocaine $P < 0.05$, thin spines sham versus cocaine $P > 0.05$; Supplementary Fig. 3a–c). These results suggest that the enhancement of the AMPA/NMDA ratio observed in D1-MSNs from cocaine-exposed mice is a result of a potentiation of AMPA receptor-mediated currents at excitatory synapses.

Potentiation at D2-MSNs only in those with negative scores

Although inputs to D2-MSNs did not show potentiation in the majority of mice, analysis of individual mice revealed that a fraction of cocaine-exposed mice showed an increase in the AMPA/NMDA ratio in D2-MSNs (Fig. 3a). Among mice with high daily cocaine intake, we found a strong negative correlation between the behavior scores

Figure 3 Synaptic potentiation at D2-MSNs correlates negatively with negative behavior scores. (**a,d**) AMPA/NMDA ratio as a function of cocaine intake in D2-MSNs (**a**) and D1-MSNs (**d**) from sham mice (gray) and all cocaine mice that underwent 17 or more self-administration sessions (D2-MSN, red, $n = 18$ mice; D1-MSN, green, $n = 22$ mice). Shaded area represents sham mean $\pm$ s.d. (**b,e**) Behavior scores as a function of AMPA/NMDA ratio in D2-MSNs ($n = 8$ mice, **b**) and D1-MSNs ($n = 15$ mice, **e**) from mice with high intake and that completed the behavioral analysis. Color lines are linear regression fit and 95% confidence interval. (**c,f**) AMPA/NMDA ratios recorded in D2-MSNs (**c**) and D1-MSNs (**f**) from sham (black, $n = 15$ mice) and cocaine self-administration mice with positive (+) and negative (–) behavior scores (D2-MSN, $n = 4$ and 4 mice; D1-MSN, $n = 8$ and 7 mice). Dotted line marks the mean AMPA/NMDA ratio in sham mice. Data are presented as mean $\pm$ s.e.m. (see **Supplementary Table 1**). * $P < 0.05$, one-way ANOVA followed by Bonferroni.



and the ratio at D2-MSN inputs ($r_p = -0.81$, $r^2 = 0.66$, $P = 0.01$, $n = 8$ mice; **Fig. 3b**). Mice with negative behavior scores displayed potentiated inputs, whereas mice with positive behavior scores had ratios that were comparable to sham controls (positive, ratio = 2.6 ± 0.3 , behavior score = 0.33 ± 0.14 , $n = 4$ mice; negative, ratio = 5 ± 1 , behavior score = -0.64 ± 0.22 , $n = 4$ mice; sham, ratio = 2.1 ± 0.9 , $n = 15$ mice; one-way ANOVA, $F_{2,20} = 9.7$, $P < 0.01$; Bonferroni *post hoc* negative versus sham, $t = 4.4$, $P < 0.05$; positive versus sham, $t = 0.6$, $P > 0.05$; **Fig. 3c**). Notably, mice with positive and negative behavior scores had similar cocaine intake (positive, 14.2 ± 1.3 mg per kg per d; negative, 14.5 ± 2.1 mg per kg per d), indicating that the dose of cocaine does not account for this difference.

No correlation was observed between D1-MSNs and the behavior scores ($r_p = -0.02$, $r^2 < 0.01$, $P = 0.95$, $n = 15$ mice; **Fig. 3d,e**). Mice with positive and negative behavior scores had similar AMPA/NMDA ratios, and synaptic potentiation in D1-MSNs was therefore not correlated with the development of compulsive cocaine intake (positive, ratio = 2.9 ± 0.7 , behavior score = 1.4 ± 0.8 , $n = 8$ mice; negative, ratio = 3.2 ± 0.5 , behavior score = -0.87 ± 0.2 , $n = 7$ mice; sham, ratio = 1.9 ± 1.1 , $n = 15$ mice; one-way ANOVA, $F_{2,20} = 2.8$, $P > 0.05$; **Fig. 3f**).

D2-MSN inhibition enhances motivation for cocaine

These findings indicate that excitatory inputs with higher AMPA/NMDA ratios are present in D2-MSNs of individuals that did not develop compulsive behaviors toward cocaine, suggesting that potentiation of excitatory inputs onto D2-MSNs might confer protection against cocaine-related behaviors. We reasoned that weakening of the indirect pathway may render individuals more vulnerable to the expression of compulsive behavior toward cocaine. To test this hypothesis, we manipulated the activity of D2-MSNs using a chemogenetic approach with designer receptor exclusively activated by a designer drug (DREADD)^{18,19}. hM4D_i is a G_i-coupled DREADD that has been used to inhibit MSN activity in the striatum^{15,20}. A conditional viral vector expressing hM4D_i was injected bilaterally in the NAc core of mice expressing Cre recombinase selectively in D2-MSNs (*Adora2a-cre^{+/+}* mice). Fluorescently tagged DREADDs were expressed at the soma of MSNs in the NAc core region around the anterior commissure and on axonal projections to the ventral pallidum (**Fig. 4a,b**). No labeling was seen in other brain regions.

The inhibitory action of hM4D_i on the indirect pathway was confirmed with recordings from ventral pallidum neurons. Synaptic responses were triggered by selective activation of fibers from D2-MSNs in the NAc that coexpressed channelrhodopsin-2 (ChR2) and hM4D_i (injected at a 1:1 ratio). A laser pulse evoked GABAergic inhibitory postsynaptic currents (IPSCs) that were depressed by $52.5 \pm 3.7\%$ when the synthetic agonist for hM4D_i, clozapine *N*-oxide (CNO, 10 μ M), was applied (paired *t* test, $t_6 = 4.0$, $P < 0.01$, $n = 7$ cells; **Fig. 4c,d**). Application of CNO had no effect on the amplitudes of IPSCs in slices expressing only ChR2 in D2-MSN (data not shown). These results indicate that activation of hM4D_i expressed in accumbal D2-MSNs inhibits the output of the accumbal-ventral pallidum indirect pathway.

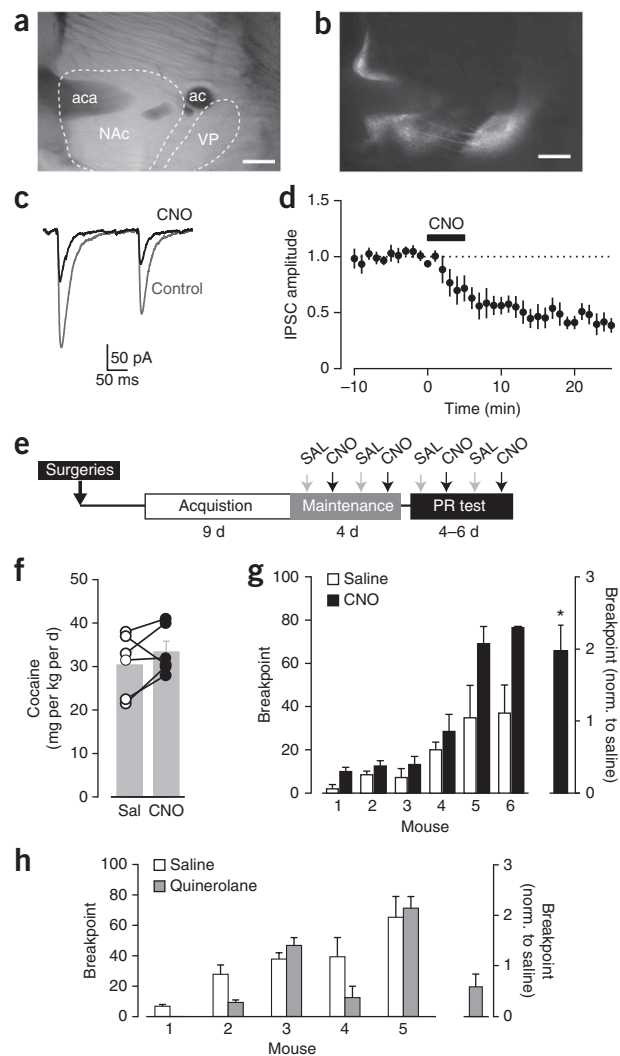
Mice expressing hM4D_i in D2-MSNs in the NAc were allowed to acquire cocaine self-administration behavior for 9 d. We first examined the effect of D2-MSN inhibition on sessions in which the drug was easily accessible under a fixed pokes:reward ratio of 1 (FR1). Mice alternately received CNO (1 mg per kg) and saline intravenously immediately before the session (**Fig. 4e**). The two groups showed similar daily cocaine intake (saline = 31 ± 2.9 mg per kg per d; CNO = 34 ± 2.3 mg per kg per d, paired *t* test, $t_{13} = 0.86$, $P = 0.24$, $n = 6$ mice; **Fig. 4f**), suggesting that inhibition of the D2-MSNs output did not alter cocaine intake when the drug was available with minimum effort. The motivation to obtain cocaine was tested in 4–6 consecutive progressive ratio sessions in which the pokes:reward ratio was increased exponentially with each subsequent reward. Notably, during these sessions, drug intake dropped to 10–40% of the normal intake during sessions with low-effort access to cocaine (FR1). Following a latin-square design, CNO or saline was administered intravenously immediately before the session. Mice had different breakpoint values on control saline days (**Fig. 4g**), which reflected the individual differences in the motivation for cocaine (**Fig. 1**). However, independently of their baseline breakpoint, mice had more active pokes and achieved higher breakpoint on days in which they were treated with CNO (breakpoint: saline = 18.7 ± 6.1 , CNO = 35.4 ± 12.3 , $n = 6$, paired *t* test, $t_5 = 2.6$, $P < 0.05$). When normalized to saline, the breakpoint doubled on CNO days, indicating a higher motivation to obtain cocaine following inhibition of D2-MSNs (normalized breakpoint in CNO = 2 ± 0.2 , one-sample *t* test, $t_5 = 2.7$, $P < 0.02$; **Fig. 4g**).

Figure 4 Inhibition of indirect-pathway D2-MSNs increases motivation for cocaine. **(a,b)** Bright field image of sagittal section **(a)** and fluorescence image **(b)** showing DREADD expression in NAc neurons and projections to ventral pallidum (VP). aca, anterior commissure ant; ac, anterior commissure. Scale bars represent 500 μ m. **(c)** Representative traces of ChR2-evoked IPSCs recorded in ventral pallidum neurons before (gray) and after CNO (10 μ M) application. **(d)** Time course of the CNO inhibition of IPSC amplitude in ventral pallidum neurons ($n = 7$). **(e)** Diagram of the experimental timeline. **(f)** Cocaine intake for mice receiving saline (open) or CNO (solid). **(g)** Left, breakpoint values achieved by an individual mouse during 4–6 consecutive progressive ratio sessions (PR test) in which mice randomly received saline (open) or CNO (solid, 1 mg per kg). Inset, breakpoint values after CNO normalized to individual saline values. $*P < 0.05$, one sample t test. **(h)** Breakpoint values achieved by an individual mouse during four consecutive progressive ratio sessions (PR test) in which mice randomly received saline (open) or the D2 agonist quinolorane (gray, 0.03 mg per kg). Inset, breakpoint after quinolorane normalized to individual saline values. Data are presented as mean $\pm$ s.e.m.

Treatment with CNO had no effect on breakpoint values in control mice that received intracranial injection, but did not express hM4Di (*Adora2a-cre*^{−/−} mice or *Adora2a-cre*^{+/−} mice with unsuccessful injections), ruling out a nonspecific effect of CNO on performance (breakpoint: saline = 41.5 ± 18.2 , CNO = 33.9 ± 16.9 , paired t test, $t_3 = 1.3$, $P > 0.2$, $n = 4$ mice; breakpoint in CNO normalized to saline = 0.78 ± 0.16 , one sample t test, $t_3 = 1.35$, $P = 0.27$; **Supplementary Fig. 4a**). When hM4Di was expressed in D2-MSNs of the dorsal striatum, CNO treatment did not increase breakpoint values, suggesting a selective regulation by NAc neurons of cocaine seeking during the first weeks after acquisition of the behavior (breakpoint: saline = 59.2 ± 24.1 , CNO = 66.8 ± 23.6 , paired t test, $t_5 = 0.5$, $P > 0.6$, $n = 6$; breakpoint in CNO normalized to saline = 1.2 ± 0.2 , one sample t test, $t_5 = 1.3$, $P > 0.2$; **Supplementary Fig. 4b,c**).

Activation of G_i signaling in D2-MSNs can also be achieved by administration of D2 receptor agonists. However, D2 receptors are also expressed in other cell types, including striatal cholinergic interneurons and midbrain dopaminergic neurons, where activation of these receptors inhibits dopamine release²¹. Systemic administration of the D2-like agonist quinolorane failed to increase breakpoint values for cocaine during progressive responding sessions (saline = 35 ± 9.5 , quinolorane = 28 ± 13.4 , $n = 5$ mice, paired t test, $t_4 = 1.1$, $P > 0.3$; **Fig. 4h**). In contrast, a trend to decreased responding for cocaine was observed in these experiments when a low dose of quinolorane (30 μ g per kg) was administered intravenously before the start of the session (normalized breakpoint in quinolorane = 0.59 ± 0.24 , one-sample t test, $t_4 = 1.67$, $P = 0.17$, $n = 5$ mice; **Fig. 4h**). This dose was chosen because higher doses of D2 receptor agonist depress locomotor activity in mice²², which would interfere with the task. This result is consistent with previous reports that D2-like agonists produce a leftward shift in the dose-response curve for cocaine self-administration and decrease overall cocaine intake in rats and nonhuman primates^{23–25}.

This same activity manipulation had no effect on responding when mice were trained on a cued-instrumental task to earn a food reward (**Supplementary Fig. 5**). *Adora2a-cre*^{+/−} mice expressing hM4Di in NAc D2-MSNs ($n = 3$ mice) and sham surgery control *Adora2a-cre*^{−/−} mice ($n = 4$ mice) were trained on a fixed ratio schedule and the effect of CNO on responding for food rewards was assessed under fixed ratio and progressive ratio schedules. Food rewards earned under fixed ratio were similar on days on which mice received saline and CNO in both genotypes (rewards per h in *Adora2a-cre*^{+/−} mice: saline = 54 ± 10 , CNO = 53 ± 11 ; *Adora2a-cre*^{+/−} mice: saline = 57 ± 6 , CNO = 54 ± 10 ; no interaction or effect in repeated-measure two-way

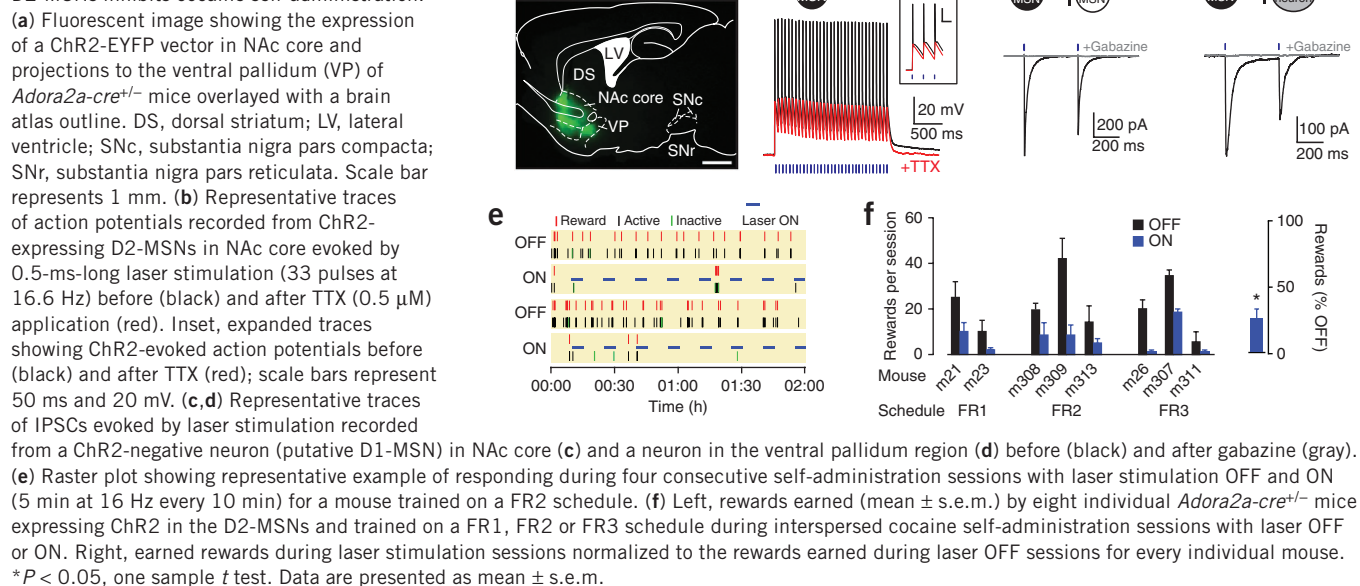


ANOVA, all F 's < 0.1 , P 's > 0.1 ; **Supplementary Fig. 5b**). When normalized to rewards earned on saline days, food rewards earned on CNO days were comparable between genotypes (*Adora2a-cre*^{+/−} mice = 1.0 ± 0.1 , *Adora2a-cre*^{−/−} mice = 1.06 ± 0.1 , unpaired t test, $t_{12} = 0.34$, $P = 0.7$; **Supplementary Fig. 5c**). CNO had no effect during progressive ratio sessions, and sham mice and mice expressing hM4Di achieved similar breakpoints for food on saline and CNO days (*Adora2a-cre*^{+/−} mice: saline = 983 ± 362 , CNO = 829 ± 277 ; *Adora2a-cre*^{−/−} mice: saline = 950 ± 153 , CNO = $1,341 \pm 388$; no interaction or effect in repeated-measure two-way ANOVA, all $F_{1,5}$'s < 0.8 , P 's > 0.4 ; **Supplementary Fig. 5d**). Normalized CNO breakpoints were similar between genotypes (*Adora2a-cre*^{+/−} mice = 0.98 ± 0.3 , *Adora2a-cre*^{−/−} mice = 1.7 ± 0.8 , unpaired t test, $t_5 < 0.98$, $P > 0.4$; **Supplementary Fig. 5e**), supporting our findings that inhibition of accumbal D2-MSNs did not affect motivation for food reinforcement. Furthermore, striatal D1 receptor expression and function appeared to be unaltered in mice that expressed hM4Di in the striatum and received repeated CNO treatment (1 mg per kg for 2–4 d; **Supplementary Fig. 6**).

D2-MSN activation suppresses cocaine self-administration

In vivo optogenetic stimulation was used to selectively activate D2-MSNs in the NAc core and test the effect of accumbal indirect pathway stimulation on cocaine self-administration. We bilaterally injected a

Figure 5 Activation of indirect-pathway D2-MSNs inhibits cocaine self-administration.



conditional viral vector expressing ChR2 (EYFP tagged) into the NAc core region of *Adora2a-cre^{+/-}* mice and implanted fiber optics in the injection area. Pathway-specific expression of ChR2 was confirmed using fluorescent microscopy, which revealed labeled cell bodies in the NAc core and labeled projections in the ventral pallidum, but not in other brain regions (Fig. 5a). Electrophysiological recordings in brain slices containing NAc core showed that brief laser stimulation (0.5-ms duration pulse at 16.6 Hz) reliably triggered the firing of action potentials in D2-MSNs expressing ChR2 and, as expected, the firing was sensitive to tetrodotoxin (TTX; $n = 2$ cells; Fig. 5b). The same brief laser stimulation elicited an IPSC in two types of neurons that are known to receive inputs from indirect pathway D2-MSNs: neighboring striatal MSNs (ChR2-negative, putative D1-MSNs, $n = 3$ cells) and ventral pallidum neurons ($n = 9$ cells; Fig. 5c,d). Application of the GABA_A receptor antagonist gabazine (5 μ M) abolished the laser-evoked IPSCs in both cell types ($n = 2$ for both cell types). These results confirm that optogenetic stimulation activates accumbal indirect pathway neurons and, in turn, inhibits the post-synaptic targets of D2-MSNs.

Mice expressing ChR2 in accumbal D2-MSNs were allowed to acquire cocaine self-administration behavior (5–9 d) and then habituated to perform with the optic fibers until responding was stable. Testing occurred over four consecutive daily sessions in which the laser power was OFF or ON, following the same pattern for all mice (OFF-ON-OFF-ON; Fig. 5e). In the first set of experiments, optogenetic stimulation (10-ms pulses at 16.6 Hz, 10 min OFF followed by 5 min ON throughout the session) was delivered to the NAc core. Under these conditions, the mean number of earned rewards dropped significantly during ON sessions for all mice (OFF rewards per session = 21.8 ± 4.3 , ON rewards per session = 7.3 ± 2.1 , paired *t* test, $t_7 = 4.5$, $P < 0.01$, $n = 8$ mice; Fig. 5f). When normalized to the OFF sessions, optogenetic stimulation suppressed cocaine self-administration by $69.1 \pm 5.3\%$ (one-sample *t* test, $t_7 = 12.6$, $P < 0.001$, $n = 8$ mice; Fig. 5f). Furthermore, stimulation resulted in a similar decrease in mice trained on a FR1, FR2 and FR3 schedule (FR1: off = 18 ± 7.5 , on = 6.5 ± 4 , $n = 2$ mice; FR2: off = 25.7 ± 8.5 , on = 7.8 ± 1.2 , $n = 3$ mice; FR3: off = 20.5 ± 8.3 , on = 7.3 ± 5.8 , $n = 3$ mice; two-way repeated-measure ANOVA main effect of stimulation, $F_{1,5} = 15.25$, $P < 0.02$; no effect

of fixed ratio schedule or interaction, both F 's < 0.3 , P 's > 0.7). These results indicate that recruitment of the indirect pathway reduces cocaine self-administration independent of the ratio requirement.

DISCUSSION

We found that resilience to compulsive cocaine use was accompanied by potentiation of glutamatergic inputs to D2-MSNs in the NAc, suggesting that this synaptic potentiation serves a protective mechanism. We challenged this conclusion by manipulating the activity of indirect-pathway D2-MSNs. Inhibition of D2-MSNs during self-administration enhanced responding and increased motivation to obtain cocaine, rendering mice more vulnerable to the expression of compulsive behaviors. In contrast, activation of D2-MSNs decreased cocaine self-administration. These experiments provide evidence of a causal link between the synaptic strength at D2-MSN in the NAc core and behavioral control over reward-motivated actions.

We also found synaptic potentiation of glutamatergic inputs onto direct-pathway D1-MSNs after cocaine self-administration. Enhanced AMPA/NMDA ratios and other evidence of synaptic potentiation have been seen in D1-MSNs of the NAc after passive cocaine administration^{26,27}. Our results add to previous findings by demonstrating that potentiation of D1-MSNs inputs is a generalized response to cocaine exposure and that it develops in the majority of the mice, independently of whether or not they show compulsive behaviors toward cocaine. In contrast, potentiation at D2-MSN inputs was observed only in the resilient individuals that did not develop compulsive behaviors. It remains unclear whether resilient individuals have stronger inputs to accumbal D2-MSNs, which precede the cocaine exposure, or whether their resilience developed as a consequence of repeated cocaine self-administration. Judging by the range of AMPA/NMDA ratios that we recorded in sham mice, it is unlikely that the potentiation of D2-MSNs inputs preceded drug exposure. However, further investigation is required to determine the predisposing factors to this resilience. Manipulations of the activity of indirect-pathway neurons in the NAc, even when introduced after several weeks of drug taking, substantially increased or decreased cocaine seeking and self-administration. Systemic administration of a D2-like agonist did not enhance seeking and failed to reproduce the effects of inhibiting

D2-MSNs via cell-specific expression of hM4Di. This is possibly a result of the different effects of activation of dopamine D2 receptors expressed on other cell types, such as dopaminergic neurons and cholinergic interneurons. These results highlight the importance of using cell-specific approaches and demonstrate that selective activation of G_i signaling in D2-MSNs can increase drug-seeking behavior. The same manipulation did not alter responding for food reward, suggesting that the effect is specific for cocaine seeking. Dorsal regions of the striatum are more likely to be involved, as they regulate cued-instrumental responding for food²⁸. As expected for a strong reinforcer, breakpoint values achieved for food were 10–30-fold higher than those achieved for cocaine. Thus, it is possible that the activity of indirect-pathway neurons is already depressed when responding for such a potent reinforcer and the effect of hM4Di activation might be occluded. When D2-MSNs in the dorsal striatum were inhibited after 3 weeks of cocaine self-administration, responding for cocaine was not increased, suggesting that indirect-pathway neurons in the dorsal striatum are not involved in cued drug seeking at this early stage and under these experimental conditions.

D2-MSNs in the NAc core send GABAergic projections to the dorsolateral subregion of the ventral pallidum, where neurons display sustained changes in firing rate that are time locked to responding during a cocaine self-administration task²⁹. Reinstatement of cocaine seeking is associated with decreased levels of extracellular GABA in the ventral pallidum, and blockers of glutamatergic transmission in the NAc core prevent the change^{30,31}. These findings implicate indirect D2-MSNs in NAc and its projections to the ventral pallidum in the processing of drug-seeking behavior and inhibitory control of the output. It is tempting to speculate that selective silencing of ventral pallidum neurons would also decrease the motivation to seek cocaine. Using therapeutic brain stimulation to enhance indirect pathway or silence the output of ventral pallidum in patients could enhance self-control in those fighting a dependence on stimulant drugs.

In conclusion, our results suggest that synaptic potentiation in D2-MSN inputs is a critical mechanism for controlling the expression of compulsive behaviors toward cocaine. We propose that this cell-specific synaptic potentiation facilitates the recruitment of indirect-pathway neurons and protects against the development of addictive behaviors.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Supplementary information is available in the [online version of the paper](#).

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AUTHOR CONTRIBUTIONS

R.B. carried out the behavioral studies with help from C.H.C., E.M., P.F.K. and C.M.G. J.H.S. carried out all of the electrophysiological experiments with help from M.F.A. A.D. performed all of the two-photon glutamate uncaging experiments. A.R.K., C.M.G., R.B. and M.F.A. performed behavioral experiments with hM4Di and *in vivo* optogenetics. R.B., J.H.S. and V.A.A. designed the study and wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Mice. Experiments were performed in accordance with guidelines from the National Institute on Alcohol Abuse and Alcoholism's Animal Care and Use Committee. Drd1-EGFP BAC transgenic mice (GENSAT, Swiss Webster background) and *Adora2a-cre* BAC transgenic mice (GENSAT, KG139Gsat/Mmcd, SW/B6 background) were used. Mice were housed in groups until surgery and individually afterwards on reversed light cycle (lights on 18:30–06:30) with food and water *ad libitum*.

Intravenous cocaine self-administration. Mice (~40 d old) were implanted with a chronically indwelling catheter (CamCath) in the right jugular vein as described previously³². Mice received oral antibiotics and recovered for 5–7 d before behavioral testing was started. Catheters were maintained by daily flushes with heparinized saline (100–275 U ml⁻¹). Sham surgeries permanently occluded blood flow through the jugular vein. Sham control mice (*n* = 20) were left undisturbed in their home cage for at least 3 weeks in the reversed light cycle before experiments. Modified operant boxes (Med-Associates) (size = 11 × 18 × 13 cm) were used and behavioral testing was performed during the dark phase of the light cycle. Pokes in active hole delivered intravenous cocaine infusions (1–2 mg per kg) paired with the cue light ON for 4 s and a 10-s time out. The house light signaled the drug-off periods and the end of the sessions. Naive mice were trained in 2–6-h sessions for 5–13 d on a fixed poke:reward ratio of 1. The criteria for acquisition of self-administration behavior was a ratio of active:inactive pokes of at least 2.5 and cocaine intake higher than 5 mg per kg per d for the last two training sessions. An estimated 55% of Swiss Webster Drd1-EGFP mice (27 males, 9 females) reached criteria within 8.5 ± 0.5 d of training and were moved to sessions (2:30 h long, cocaine = 2 mg per kg per infusion) and self-administered cocaine for an average of 31 more days (min = 17, max = 41; **Supplementary Table 1**). The other 45% of the mice (31 mice) did not acquire self-administration behavior. The behavioral analysis was completed in 28 out of the 36 mice. 8 of 36 mice lost catheter patency right before the last breakpoint session and were used for physiological recordings, but lacked behavioral measurements (**Supplementary Table 1**). Sessions consisted of three drug-on periods (40 min each) interspersed by two 15-min drug-off periods. Perseverance was measured as the number of active pokes during the drug-off periods over the last five sessions. Motivation was measured in two exponentially progressing responding sessions (after days 15 and 30) and the breakpoint was assigned as the motivation value³³. The motivation and perseverance scores for individual mice were calculated using the *z* score,

$$\frac{X_i - \bar{X}}{\text{s.d.}}$$

where X_i is the behavior value for the individual mouse, $\bar{X}$ is the mean behavior value for all of the mice included in the study, and s.d. is the standard deviation of the population. The combined behavior score was equal to the sum of the motivation and perseverance scores for each individual mouse.

In experiments using *Adora2a-cre*^{+/+} mice, sessions were 3:40 h long, the single infusion dose was 1 mg per kg of cocaine, and mice were trained on a fixed ratio of 1, 2 or 3 as mentioned and depending on the experiment. CNO and quinelorane were dissolved in saline and administered intravenously (1 ml kg⁻¹) at doses of 1 mg per kg and 0.03 mg per kg, respectively, through the catheter right before the start of the session. Catheter patency was determined every 10 d by anesthetic cocktail (1.5 mg per kg ketamine, 0.75 mg ml⁻¹ midazepan). Mice that failed patency before reaching session 25 were removed from the study. Behavioral data was analyzed and plotted in Igor Pro. The log normal equation

$$y = y_0 + A e^{\left(\frac{\ln \frac{x}{x_0}}{\text{width}} \right)^2}$$

was used to fit the behavioral values.

Intracranial viral gene transfer and *in vivo* optogenetic stimulation. Cre-inducible AAV-hSyn-DIO-hM4Di-mCherry (serotype 1 or 2, 6 × 10¹² virus molecules per ml, Optogenetics and Transgenic Technology Core at the National Institute on Drug Abuse and Gene Therapy Vector Core at University of North Carolina) and AAV5-EF1a-DIO-ChR2(H134R)-EYFP (4 × 10¹² VM ml⁻¹, UNC Vector Core)

were used. Stereotaxic bilateral injections (250–400 nl at 100 nl min⁻¹) were performed into the NAc core (anterior-posterior, +1.4 mm; medial-lateral, ±0.1 mm; dorsal-ventral, –4.8 mm, from Bregma) or the dorsal striatum (anterior-posterior, +0.8 mm; medial-lateral, ±1.75 mm; dorsal-ventral, –3.2 mm, from Bregma) of *Adora2a-cre*^{+/+} and *Adora2a-cre*^{-/-} mice at 5–6 weeks of age. For self-administration experiments, mice recovered for 3–5 d before indwelling catheters were implanted in the jugular vein. To confirm hM4Di activity using electrophysiological recordings, we injected both Cre-inducible hM4Di-mCherry and ChR2-EYFP vectors (1:1) into NAc core of *Adora2a-cre*^{+/+}. For *in vivo* optogenetic stimulation of indirect pathway, a two-ferrule cannula with fiber optics (200 μm/0.22 NA, 4.0 mm, Doric Lenses) was implanted immediately after the viral injection right above the injection area and secured to the skull with metabond and dental cement. Diode-pumped blue lasers (473 nm, 25 mW, CrystaLaser) was used and connected via optic fiber (200 μm/0.22 NA, M25L02, ThorLabs) to a 1in-2out fiberoptic rotary joint (Doric Lenses) to split laser power and facilitate movement during the task. Stimulation consisted of 10-ms pulses delivered at 16.6 Hz for 5 min (2 min for one mouse), mean power output of 5 mW (3.5–6 mW) at each fiber tip of the two-ferrule cannula implanted in the NAc core every 10 min throughout the session. The control mice for these experiments were *Adora2a-cre*^{+/+} mice injected with AAV-hSyn-DIO-mCherry (serotype 5, titer = 6 × 10¹² virus molecules per ml, Gene Therapy Vector Core at University of North Carolina).

Electrophysiology. Sagittal slices (250 μm) were prepared from 103 ± 12-d-old mice (*n* = 35) 4–17 d after last self-administration session and from sham-surgery mice (97 ± 16 d old, *n* = 20) in cutting solution containing 225 mM sucrose, 119 mM NaCl, 2.5 mM KCl, 0.1 mM CaCl₂, 4.9 mM MgCl₂, 26.2 mM NaHCO₃, 1 mM NaH₂PO₄, 1.25 mM glucose, and 3 mM kynurenic acid²⁶. Slices were incubated at 33 °C for 30 min and at 22–24 °C until used. Recordings were made at 30 °C using artificial cerebrospinal fluid (ACSF) containing 124 mM NaCl, 2.5 mM KCl, 2.5 mM CaCl₂, 1.3 mM MgCl₂, 26.2 mM NaHCO₃, 1 mM NaH₂PO₄ and 20 mM glucose. Gabazine (5 μM) and D-serine (10 μM) were added to the ACSF to isolate glutamatergic EPSCs. Whole-cell voltage-clamp recordings were obtained from GFP- positive and GFP-negative MSNs in the NAc core using electrodes (2.5–3.5 MΩ) filled with solution containing 130 mM CsMeSO₄, 10 mM CsCl, 10 mM HEPES, 0.2 mM EGTA, 4 mM Na-ATP, 0.4 mM Na-GTP and 0.1 mM spermine (pH = 7.25, ~310 mOsm). Recordings were excluded if access resistance was greater than 25 MΩ or if input resistance was greater than the range expected for MSNs (>350 MΩ). eEPSCs were elicited by a pair of current pulses (0.2-ms width, 50-ms interval, every 20 s) using an ACSF-filled glass pipette placed ~250 μm rostro-dorsally. AMPA receptor-mediated eEPSCs were recorded when holding cells at –70 mV. eEPSCs were then recorded at +40 mV in the presence or absence of the NMDA receptor antagonist (R)-CPP (10 μM) and the NMDA receptor response was calculated by subtraction off-line. For current-clamp recording from D2-MSNs, potassium-based internal solution was used. Measurements of D2-MSN output in either core or the ventral pallidum were carried out by whole-cell voltage-clamp recordings using CsCl-based internal solution (all other components were kept the same as described above) while holding the cells at –20 mV in the presence of the AMPA receptor antagonist NBQX (5 μM) to isolate GABAergic IPSCs. IPSCs were evoked by activation of ChR2 and triggered by a brief light pulse (0.2-ms width, 20-s interval) produced by a diode-pumped blue laser (473 nm, 25 mW, CrystaLaser) and delivered via a fiber optic (200 μm/0.22 NA, ThorLabs). All data was analyzed and plotted in Igor. CNO stock was prepared in water and tested at a final concentration of 10 μM in ACSF. Data was acquired using Multiclamp 700B (Molecular Devices), filtered at 1 kHz and digitized at 5 kHz. Electrophysiological recordings were made blind to the behavior score of the mice, except for sham-surgery mice that did not carry an intravenous port and were easily distinguishable to the experimenter. Data was collected from 16 sham-surgery mice and 27 mice after cocaine self-administration within 1–38 d of the last cocaine exposure. Behavioral data was available for 22 mice (5 mice had no behavior; **Supplementary Table 1**). Recordings from D1-MSNs were collected in 9 mice, from D2-MSNs in 3 mice, and from D1-MSNs and D2-MSNs in 14 mice. Note that there is no behavioral data for all of them and refer to **Supplementary Table 1** for detail.

Two photon glutamate uncaging. A custom-built two-photon laser-scanning microscope equipped with two Ti:Sapphire lasers was used to simultaneously image dendritic spines using a 840-nm laser and photo-release caged glutamate

next to the spines using a 720-nm laser. Slices and solutions were prepared as described above. Whole-cell recordings were obtained from GFP-positive MSNs in the NAc core region at 25 °C. Alexa Fluor 594 (10 μ M) was added to the internal solution to visualize second- and third-order dendrites and spines within 200 μ m of the soma were randomly selected for imaging. For all experiments, MNI-glutamate (5 mM), GABA_A (5 μ M), TTX (1 μ M) and the Ca²⁺ channel blockers mibefradil (20 μ M) and conotoxin GVIA (1 μ M) were added to ACSF. For recordings of AMPA receptor uEPSCs, cells were held at -80 mV and AP5 (5 μ M) was added to the external solution. For recordings of NMDA receptor uEPSCs, cells were held at -50 mV and low Mg²⁺ (50 μ M) ACSF containing NBQX (5 μ M) was used. Glutamate uncaging was achieved with a 1–2-ms pulse of 720-nm light delivered next to the spine head every 30 s, and the laser power was standardized in advance by measuring the fluorescence recovery from photobleaching of the red dye in the spine head³⁴. Data were acquired using National Instruments data acquisition boards and custom software written in MATLAB³⁵. We averaged 10–15 uEPSCs per spine and calculated the peak amplitude in a 1-ms window around the peak. Recordings were collected from D1-MSNs in three cocaine mice and four sham-surgery mice (Supplementary Table 1).

Operant food self-administration. *Adora2a-cre^{+/−}* and sham *Adora2a-cre^{−/−}* littermate mice were stereotactically injected into NAc core with Cre-inducible AAV vectors expressing hM4Di-mCherry and allowed to recover for 9 d. Mice were placed in operant chambers in sound-attenuating boxes (Med Associates) in which they pressed a cued-active lever (left or right, counterbalanced across mice) for an outcome of regular chow pellets (20-mg pellets). Presses on an inactive lever did not result in reinforcement. Mice were food restricted 2 d before training commenced to 90% of their baseline weight, which they were maintained at for the duration of the experiments. On the first day, mice were trained to approach the food magazine (no levers or cue light present) on a random time schedule, with a reinforcer delivered, on average, every 60 s for a total of 30 min. All other training sessions commenced with illumination of the house light, active and inactive lever extension, and illumination of a cue light located directly above the active lever, and ended after 60 min with the levers retracting and the cue and house light turning off. Mice were trained on a FR1 schedule for 3–5 d and, once lever-pressing behavior was acquired, the fixed ratio requirement was increased to FR10. One hour before each session, mice received an injection of saline (10 ml kg^{−1}, intraperitoneal) during training for habituation and they received either saline or CNO (1 mg per kg dose, 10 ml kg^{−1}, intraperitoneal) during testing. Testing of FR10 responding began after 3 d of baseline FR10 responding and consisted of 2 d of CNO and 2 d of saline injections. The order of drug test (saline first versus CNO first) was counterbalanced across mice and kept constant across CNO testing. Progressive ratio testing followed in which mice were given CNO or saline on alternating days (counterbalanced across mice) and with a FR10 training session in between progressive ratio test days. The same progressive ratio schedule used for cocaine self-administration was used here³³ with an initial breakpoint of 10, and maximum breakpoint to be achieved of 1012.

Locomotor assessment. Locomotor activity was recorded in cages (10.25 × 6 inches) constructed out of clear polycarbonate walls and floors under dim illumination (150 lx). Horizontal activity was detected as infrared beam crosses (1-inch spacing, ten beams per cage) made on consecutive beams (ambulatory counts) using Opto M3 activity monitors (Columbus Instruments). The response to the D1 agonist SKF81297 was assessed using a previously described protocol²². Briefly, adult mice (8–16 weeks old) were allowed to run freely for 60 min, during

which baseline locomotor activity was determined. They received an intraperitoneal injection of either saline or 2.5 mg per kg SKF81297, a dose that increases locomotor activity in wild-type C57BL/6J mice³⁶, were immediately returned to the cage and horizontal locomotion was measured for the next 150 min. In all cases, locomotor activity is expressed as the number of ambulatory counts per 10 min, 1 h or normalized to the baseline value obtained for each mouse before drug administration.

Quantitative PCR. Striatal tissue was microdissected from two adult mice (9–16 weeks old) of each genotype (*Adora2a-cre^{+/−}* and control *Adora2a-cre^{−/−}*). All mice received an intra-striatal injection of a viral vector that conditionally expresses hM4Di and 2–4 d of CNO (1 mg per kg) treatment. Total RNA was purified using RNeasy Micro (Qiagen), and cDNA was synthesized using iScript cDNA Synthesis Kit (Bio-Rad). mRNA expression of dopamine D1 receptor *Drd1a* and the endogenous control gene *Actb* were determined using TaqMan Gene Expression Assays (Applied Biosystems). Quantitative PCR runs were performed using TaqMan Fast Polymerase (Applied Biosystems) in a StepOnePlus Real-Time PCR system. Cycling conditions were as follows: initial hold at 95 °C for 20 s, 40 cycles of step 1 (95 °C for 1 s) and set 2 (60 °C for 20 s). Samples were run in quadruplicates, and negative controls were run in parallel. cDNA synthesis and quantitative PCR experiments were repeated three times. Relative quantification was calculated using the $\Delta\Delta$ Ct method (StepOne System Software, Applied Biosystems).

Statistics. All statistical comparisons were two-sided and were performed with IGOR or Prism. Paired and unpaired Student's *t* tests, one-way and repeated-measures two-way ANOVA were used when appropriate. Bonferroni tests were used for multiple comparisons *post hoc* analysis. One sample *t* test was used to compare normalized breakpoint values of all rewards earned when mentioned. *F* test was used to compare the variance among groups and test the assumption of normality for each statistical test. In the case of AMPA/NMDA ratios and uncaged-evoked responses at single spines (uEPSCs), the normality assumption was not met and the nonparametric Mann-Whitney test was used instead. All replicates were biological and the errors were calculated as s.e.m. Sample size was not justified with statistical power.

Drugs. Cocaine-HCL was obtained from the National Institute on Drug Abuse. Gabazine (SR-95531), (*R*)-CPP, NBQX, AP5, kynurenic acid and conotoxin GVIA were purchased from Ascent Scientific, MNI-glutamate, mibefradil and quinelorane from Tocris, Alexa Fluor 594 from Molecular Probes, and CNO and all other chemicals from Sigma.

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